

More female students enter traditional engineering courses

The share of female students in mechanical, civil, mining, and marine engineering undergraduate courses rose notably in the recent decade

DATA POINT

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The number of students enrolled in undergraduate engineering courses in India steadily declined from 2016, despite the increase in the overall number of students pursuing college education, according to the All India Survey on Higher Education (AISHE). In recent years, the number of students enrolled in India was highest in the arts, science, and commerce streams, followed by engineering. Engineering was also the only discipline in which the number of students enrolled consistently declined between 2016 and 2022. There was a slight uptick in 2022 in enrollment in undergraduate engineering courses, which may have been due to the introduction of several new courses in the survey, which were previously not tracked by AISHE. This bump did not alter the trend, however.

The decline in enrolment in engineering courses was more pronounced among male students than female. **Chart 1** shows the number of male and female students who enrolled for undergraduate engineering courses between FY13 and FY22.

Chart 2 shows the number of female students enrolled in specific undergraduate engineering courses and their share in the overall intake, between FY13 and FY22. In 2013, large numbers of female students took up electronics engineering, computer engineering, and information technology (IT). In computer engineering and information technology, female students comprised more than 45% of all students in 2013. As the decade progressed, the presence of female students in these disciplines dropped by some 10% points each. However, with a possible sway in societal attitudes and the intro-

duction of new courses, female students started enrolling in disciplines where they were previously had little presence. For instance, in mining engineering, the share of female students was 1.2% in 2013 and increased to 8% in 2022. Similarly, in marine engineering, the share of female students jumped from 1.5% to 6.4%. The share of female students also increased significantly in mechanical engineering and civil engineering, from 4.2% to 7.6% and 18.2% to 23%, respectively, in the same period.

Male students continued to dominate the field of mechanical engineering, with their share declining only slightly from 95.8% in 2013 to 92.4% in 2022. The absolute number of students enrolling in the field dropped significantly though, almost halving from over 10 lakh in 2016 to 5.4 lakh in 2022. On the other hand, there was a drastic increase in the number of male students enrolling in computer engineering courses, from about 3.5 lakh in 2013 to 8.25 lakh in 2022.

Chart 3 shows the share of student enrolments in undergraduate engineering courses from Scheduled Castes (SCs), Scheduled Tribes (STs), and Other Backward Classes (OBCs). The share of OBC students who enrolled in these courses steadily rose from 33% in 2013 to 38% in 2022. The share of ST students increased slightly from 2.1% to 3.3% in the same period (**Chart 3**). Among SC students, the increase was inconsistent. It slightly declined during some years, settling at 17% in 2022 compared with 8.7% in 2013. The share of students from the general category dropped over time.

Mechanical and civil engineering courses seem to have lost their appeal over the years among men, pointing towards a decline in employability, while courses such as computer engineering have continued to draw students.

Inputs from Muhammad Usbullah A. L. and Mohamed Akhavan A. L.

Defying stereotypes

The data for the charts are sourced from the All India Survey on Higher Education



Chart 1: The number of male and female students who enrolled for undergraduate engineering courses between FY13 and FY22



Chart 3: Enrolment share (%) of various communities in undergraduate engineering courses

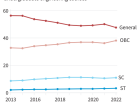


Chart 2: The chart shows the number of female students enrolled in specific undergraduate engineering courses and their share (%) in the overall intake, between FY13 and FY22

